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FORMALIZING THE ISOLATION LEMMA IN ISABELLE/HOL

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ABSTRACT

Randomized algorithms are a cornerstone of modern theoretical computer science, yet their proofs often rely on informal probabilistic intuition that obscures critical assumptions about independence and measurability. This report presents a formal verification of the **Isolation Lemma** – a fundamental principle ensuring the uniqueness of minimum-weight structures – within the **Isabelle/HOL** interactive theorem prover.

We explicitly construct the underlying product probability space for random weight assignments and formalize the conditions for the existence and uniqueness of minimizers over finite sets. By rigorously defining the auxiliary value $\alpha(x)$ and proving its independence from individual weights $w(x)$, we formally derive the upper bound on the failure probability. Furthermore, this formalization serves as the theoretical foundation for verifying a randomized parallel algorithm for the **Perfect Matching problem**. The results demonstrate how pseudo-formal probabilistic arguments can be transformed into fully machine-checked proofs, clarifying hidden dependencies in algorithmic reasoning.

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1 INTRODUCTION

Randomization has become one of the central paradigms of modern theoretical computer science. Over the past few decades, randomized methods have profoundly influenced the design and analysis of algorithms, enabling solutions that are simpler, faster, and even provably unattainable by deterministic methods.

From parallel computation to combinatorial optimization and approximation algorithms, randomness is often treated as a significant resource rather than a secondary heuristic.

Meanwhile, many core results of randomized algorithm theory are supported by probabilistic lemmas, the proofs of which are often highly abstracted. These proofs often rely on intuitive statements about probability, independence, and minimization, where detailed assumptions are often treated implicitly, and intermediate steps are frequently omitted.

While these proofs are understandable to humans, they present a significant challenge for formal verification within a rigorous logical framework, precisely because they rely on informal probabilistic intuition. In particular, probabilistic arguments often compress multiple layers of reasoning into concise informal claims, making hidden dependencies difficult to detect. This gap between informal reasoning and formal justification becomes especially pronounced in the context of randomized algorithms, where correctness is inherently probabilistic rather than absolute.

From a proof-methodological perspective, this reflects a transition from informal pen-and-paper arguments to pseudo-formal proofs that appear rigorous but still depend on unstated assumptions, and ultimately to fully formal, machine-checkable proofs. Randomized algorithm proofs are particularly sensitive to this transition, since properties such as independence, measurability, and uniqueness must be stated explicitly rather than justifies by intuition. The Isolation Lemma provides a compact yet representative case study for this transition, as it brings together these core ingredients in a single probabilistic argument.

Interactive theorem proving offers a disciplined way to address these problems. By

encoding mathematical definitions and proofs into a formal logical system and verifying them mechanically through explicit reasoning steps, theorem provers provide a level of precision and reliability that cannot be achieved solely through informal reasoning.

In recent years, there has been significant progress in the formalization of mathematics and deterministic algorithms. However, formal proofs of randomized algorithms remain relatively underexplored, largely due to the additional complexity introduced by probabilistic reasoning.

This thesis lies at the intersection of these two research directions. Its goal is to formally verify a crucial probabilistic lemma for randomized algorithms — the Isolation Lemma — in the Isabelle/HOL theorem proving system.

The Isolation Lemma is a fundamental tool in many randomized algorithms, notably in randomized algorithms for the Perfect Matching problem, where it is used to guarantee the uniqueness of minimum-weight structures with high probability. Despite its apparent simplicity, the Isolation Lemma involves several implicit assumptions about minimization, independence, and probability spaces. Explicitly stating and systematically verifying these assumptions is not only a technical challenge but also a valuable opportunity to clarify the logical structure underlying randomized algorithmic reasoning.

Through a complete formalization of the Isolation Lemma and its application to a randomized Perfect Matching algorithm, this work aims to contribute to the broader effort of bringing probabilistic algorithm theory within the scope of machine-checked formal verification.

1.1 Background

1.1.1 Randomized Algorithms in TCS

Randomized Algorithms play a fundamental role in theoretical computer science.

Unlike deterministic algorithms, randomized algorithms incorporate randomness to

break symmetry, simplify theoretical constructions, and achieve improved expected performance. Such algorithms have been widely used in matching problems, parallel computing, approximation algorithms, and the construction of complex combinatorial structures.

In many classical results, the introduction of randomized weights serves as a general technique for enforcing certain ”uniqueness” or ”isolation” properties. For example, in several parallel algorithms and combinatorial optimization problems, algorithmic correctness relies on the fact that, after a random perturbation, a minimum structure becomes unique with high probability.

The Isolation Lemma is a canonical instance of this principle. Informally, this lemma states that when random weights are assigned independently to elements of a finite universe, there is a high probability that a unique minimum-weight set exists within a given family of subsets. This lemma provides a theoretical foundation for numerous randomized algorithms and plays a critical role in algorithmic frameworks for problems such as Perfect Matching.

1.1.2 Formalization and Theorem Proving

In theoretical computer science, many fundamental results rely on intricate mathematical arguments and precise logical reasoning. As theoretical frameworks continue to develop, the scale and complexity of such proofs have grown substantially, making traditional human-based verification increasingly difficult.

Formalized mathematics and interactive theorem proving (ITP) offer a different approach to reasoning about mathematical theories. In this setting, definitions, theorems, and proofs are encoded within a formal logical system, and every inference step is checked rigorously by a computer. This provides a level of precision and reliability that is difficult to achieve through informal reasoning alone.

Compared to traditional pen-and-paper proofs, formalized theorem proving exhibits two distinctive characteristics. First, all implicit assumptions must be made explicit. These include, for example, the non-emptiness and finiteness of sets, the totality of functions, and the measurability of probability spaces. Second, the correctness of a

proof does not rely on the intuition of the reader, but is guaranteed by the underlying logical framework of the theorem proving system.

In recent years, formalized methods have been successfully applied across many areas in computer science, including program verification, compiler correctness, and the formalization of foundational mathematical theories. Within theoretical computer science, formal verification has gradually expanded from classical mathematical results to core theorems in algorithm theory and computational complexity.

However, compared to deterministic settings, the formal verification of randomized algorithms poses additional challenges. Randomized algorithms involve not only complex combinatorial structures, but also probabilistic concepts such as independence, distributions, and expectations. In a formal setting, these notions must be expressed through explicit constructions of probability spaces, rather than being assumed implicitly.

Against this background, the formal verification of key probabilistic lemmas in randomized algorithms has become an important research direction in interactive theorem proving. The Isolation Lemma serves as a representative example of such results, combining combinatorial reasoning with probabilistic arguments, and thus provides an ideal case study for formalization.

1.2 MOTIVATION

Although randomized algorithms have achieved significant success in both theory and practice, their correctness often relies on informal probabilistic and combinatorial reasoning. In traditional pen-and-paper proofs, conclusions such as “a minimum obviously exists”, “this event occurs with probability at most 1/N”, or “the relevant events are independent” are frequently treated as self-evident and left implicit.

In a formal proof setting, however, such assumptions cannot be taken for granted. All logical dependencies must be stated explicitly and verified. For instance, the existence of a minimum depends on the non-emptiness and finiteness of the underlying set; probability statements require measurability; and independence cannot be asserted

informally, but must be derived from the explicit construction of an underlying probability space. From a proof-theoretic perspective, this difficulty reflects a broader transition in proof development. Many arguments in randomized algorithms originate as informal pen-and-paper proofs, and are later refined into pseudo-formal proofs that closely resemble formal reasoning but still rely on implicit assumptions. Bridging this gap requires transforming such pseudo-formal arguments into fully formal proofs in which all probabilistic and structural dependencies are made explicit.

As a consequence, randomized algorithms and their associated lemmas constitute a particularly challenging class of objects for interactive theorem proving (ITP). On the one hand, they occupy a central position in theoretical computer science; on the other hand, they clearly demonstrate the strength of formal verification in clarifying hidden assumptions and exposing implicit reasoning steps.

The Isolation Lemma is particularly well suited to serve as a case study for this transition. Its proof simultaneously involves probabilistic modeling, independence assumptions, minimization over finite structures, and the emergence of uniqueness through randomization. As a result, formalizing the Isolation Lemma requires addressing many of the central challenges that arise when translating informal probabilistic reasoning into a fully formal proof.

This project is motivated by these considerations and aims to reconstruct the complete proof of the Isolation Lemma within a fully formalized and rigorous framework.

1.3 Project Objectives

The primary objective of this project is to formally model and prove Isolation Lemma within the Isabelle/HOL theorem proving system. Beyond establishing the statement of the lemma and its probability bound; this project aims to demonstrate a systematic transition from informal and pseudo-formal probabilistic arguments to fully formal, machine-checkable proofs.

In particular, this involves making explicit the assumptions and constructions that are typically left implicit in pen-and-paper proofs, including the underlying probability

space, independence of random variables, and minimization over finite structures. Through this process, the project seeks to illustrate how pseudo-formal reasoning formed by translation from pen-and-paper proof can be transformed into a rigorous formal proof.

Specifically, this project aims to:

- formalize random weight functions together with their corresponding product probability spaces, thereby making probabilistic assumptions explicit;
- provide a complete formal proof of the Isolation Lemma, including an explicit upper bound on the probability of non-uniqueness, and a precise characterization of the conditions under which uniqueness fails;
- apply the formalized lemma to the correctness analysis of randomized algorithms, illustrating how the resulting formal proof support higher-level algorithmic reasoning.

As a concrete application, this project employs the Isolation Lemma to formally verify a key probabilistic argument in a randomized algorithm for the Perfect Matching problem. This application serves to demonstrate the practical relevance of the formalization and show how isolation-based arguments can be carried through the full transition from informal reasoning to formal verification.

1.4 Scope of Work

The scope of this project is primarily focused on theoretical formalization and proof construction. In particular, this report covers:

- the mathematical formulation of the Isolation Lemma and its formal expression;
- the probabilistic modeling of random weight assignments and their associated properties;
- the formalization of critical construction steps within the proof;
- a representative application of the lemma to Perfect Matching.

This work does not address general derandomization techniques for randomized algorithms, nor does it consider implementation-level concerns or performance optimization of specific algorithms. Moreover, the internal implementation details of the Isabelle/HOL system and its underlying logical foundations are beyond the scope of this report.

2 PRELIMINARIES

2.1 INTERACTIVE THEOREM PROVING

2.2 ISABELLE/HOL SYSTEM

2.3 THE ISOLATION LEMMA

The Isolation Lemma, originally introduced by Mulmuley, Vazirani, and Vazirani [1], is a probabilistic principle used to ensure that, with high probability, a unique solution exists among a family of sets. It has found widespread applications in randomized algorithms, particularly in problems involving combinatorial search, matching, and derandomization.

Informal Statement. Let U be a finite set (called the *universe*), and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets of U. Each element $x \in U$ is assigned a weight $w(x)$ independently and uniformly at random from $\{1, 2, \ldots, N\}$ for some integer $N \geq 1$. The total weight of a set $S \in \mathscr{F}$ is defined as"

$$w(S) = \sum_{x \in S} w(x)$$

The Isolation Lemma states that, with high probability, there is a unique set in $\mathscr{F}$ that minimizes this weight function.

Theorem (Isolation Lemma) [1]: *Let U be a finite set with $|U| = n$, and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets. Suppose that each $x \in U$ is assigned an integer weight $w(x)$ drawn independently and uniformly at random from $\{1, 2, \ldots, N\}$. Then the probability that there exists a unique set $A \in \mathscr{F}$ that minimizers $w(S)$ is at least $1 - \frac{n}{N}$.*

$$\Pr[\exists! A \in \mathscr{F} \text{ minimizing } w(A)] \geq 1 - \frac{n}{N}$$

Proof Idea. The proof relies on analyzing the probability that *two or more* sets achieving the same minimum weight. The key technique is to consider the structure of conflicting pairs and show that their probability of collision is small. By summing over all such pairs and bounding their contribution, one obtains the upperbound $\frac{n}{N}$ on the failure probability.

Proof (Joel Spencer [2]). Let $U = \{e_1, e_2, \ldots, e_n\}$ be a finite set of n elements, and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subset of U. Suppose each element $x \in U$ is assigned a random weight $w(x)$ uniformly from $\{1, 2, \ldots, N\}$.

For any $x \in U$, define:

$$\alpha(x) = \min_{\substack{S \in \mathscr{F} \\ x \notin S}} w(S) - \min_{\substack{S \in \mathscr{F} \\ x \in S}} w(S \setminus \{x\})$$

This value $\alpha(x)$ depends only on the weights of elements other than x itself, since x is excluded from all weight sums in the definition. Because $w(x)$ is independently and uniformly chosen from $\{1, 2, \ldots, N\}$, the probability that $w(x) = \alpha(x)$ is at most $1/N$.

Now, suppose there are two distinct sets $A, B \in \mathscr{F}$ such that both minimize the total weight $w(S) = \sum_{x \in S} w(x)$. Let $x \in A \triangle B$ (i.e., an element in the symmetric difference). Without loss of generality, suppose $x \in A \setminus B$. Then we have:

$$\alpha(x) = \min_{\substack{S \in \mathscr{F} \\ x \notin S}} w(S) - \min_{\substack{S \in \mathscr{F} \\ x \in S}} w(S \setminus \{x\}) \quad = w(B) - (w(A) - w(x)) = w(x)$$

So for this x, $w(x) = \alpha(x)$. The chance that this happens for any particular x is at most $1/N$, and since there are at most n such x, the total probability that some x satisfies this is at most n/N.

Thus, with probability at least $1 - \frac{n}{N}$, no such x exists, and the minimum-weight set in $\mathscr{F}$ is unique.

Remarks.

- The bound $1 - \frac{n}{N}$ improves as N increases. In particular, for $N > 2n$, the

probability of uniqueness exceeds 0.5.

- The lemma is non-constructive: it guarantees uniqueness probabilistically, but not which set is unique.
- It is widely used in randomized parallel algorithms and in complexity theory, e.g., in the isolation of perfect matchings.

Commentary. While the above proof [2] is concise and elegant, it omits many low-level details that are often assumed in traditional mathematical exposition. For example, the definition of $\alpha(x)$ implicitly assumes that the minima involved are well-defined, and that comparisons among set weights are meaningful under all random assignments. Additionally, the reasoning about probabilities (e.g., the independence and uniformity of weights) relies on intuition rather than formal measure-theoretic grounding. These gaps, though acceptable in informal proofs, present significant challenges when one attempts to formalize such arguments in a theorem prover like Isabelle/HOL. Thus, part of the motivation of this project is bridge this gap by giving a fully rigorous, machine-checked account of the Isolation Lemma, including all technical conditions and edge cases often left implicit in classical proofs. In the next chapter, we will present a formalization of this result in Isabelle/HOL, making all implicit assumptions explicit and verifying each step within a constructive logical framework.

2.4 Parallel Algorithm for Perfect Matching

3 Formal Modeling of the Isolation Lemma

This chapter marks the first step in the formalization process by translating the core components of the Isolation Lemma into rigorous definitions with the Isabelle/HOL theorem proving environment. Building on the informal statement of the lemma provided in Chapter 2, our objective here is to move from intuitive, pen-and-paper reasoning to a precise and structured formal model. This transition corresponds to the intermediate stage in the formalization pipline: from Pen-and-Paper Proofs (PP) to a Formal Proof (FP) representation that is is machine-readable and logically well-formed, but not yet fully proved.

We proceed by constructing the essential components of the lemma layer by layer. First, we define the underlying universe of the elements and the weight functions, and explain the rationale of choosing natural numbers for weight representation. Second, we formalize the concept of the minimizer and the condition of its uniqueness within a family of sets. Finally, we construct the product probability space using HOL-Probability Library, ensuring that the weights are distributed uniformly and independently.

These definitions act as the foundation upon with the correctness proof in Chapter 4 is built. Special attention is given to the design decisions that facilitate proof automation. In particular, we carefully handle finiteness assumptions and the construction of measure spaces, as these aspects are crucial for enabling automated reasoning in Isabelle/HOL.

Standing assumptions. Throughout this chapter, we fix a universe U and a non-empty family of sets $F \subseteq \mathscr{P}(U)$, and assume that U is finite. For the randomized construction, we fix an integer $N \geq 1$ and sample weights for all $x \in U$ independently and uniformly from $\{1, \ldots, N\}$.

3.1 Modeling Weights and Universe

Informal descriptions of randomized constructions often begin with intuitive assumptions about sets and weights, relying on natural notation without explicitly encoding their constraints. In the context of formal verification, however, these assumptions must be carefully reconstructed to support mechanized reasoning.

In the standard mathematical formulation of the Isolation Lemma, we consider a universe of elements U. A weight function, w, assigns a numerical value to each element in U. Typically, these weights are integers chosen uniformly and randomly from a range $\{1,\ldots,N\}$. The weight of a subset $S \subseteq U$ is then defined as the sum of the weights of its constituent elements:

$$w(S) = \sum_{x \in S} w(x) \tag{3.1}$$

To formalize this in Isabelle/HOL, we avoid defining an explicit "Universe" type. Instead, we represent the universe implicitly via type variables $'a$. More precisely, the universe is modeled as an explicit set $U ::' a$ `set`. The type variable $'a$ serves only as a carrier type, while all semantic statements about weights are restricted to elements $x \in U$. Although weight functions are total by definition in Isabelle, values outside the universe do not play any role in the proof. We introduce a type synonym to model the weight assignment. This approach enhances the readability of the subsequent definitions, and ensures a type safety within the proof, which is necessary for the proof assistant.

```
type_synonym 'a weight = "'a ⇒ nat"
  (* weight maps an element to a natural number *)

definition wt :: "'a weight ⇒ 'a set ⇒ nat"  where
  "wt w S = (SUM x ∈ S. w x)"
  (* sum of weight *)
```

Listing 3.1: Definition of Weight Type and Weight Function

As shown in Listing 3.1, we define the type `'a weight` as a function mapping an element of an arbitrary type `'a` to a natural number (`nat`). Based on this type, the total weight of a set S is defined using the `wt` function.

The choice of a total function is a technical requirement of Isabelle/HOL, where all functions must be defined on the entire carrier type. Semantically, all arguments and constructions in the sequel will only depend on the values of w restricted to the universe U. This separation between the carrier type and the explicit universe avoids treading the weight function as a partial function, which is not natively supported in Isabelle/HOL.

This modeling step illustrates a typical transition from pen-and-paper reasoning to formal verification. While informal proofs treat the definition of weights and subset sums as self-evident, formalization forces us to separate type-level carriers from explicitly quantified finite sets, and to express totality and domain restrictions precisely. These distinctions are not only technical, but crucial in enabling mechanized proof construction within a system like Isabelle/HOL.

DESIGN DECISION: NATURAL NUMBERS VS. INTEGERS

A critical modeling decision was the choice of the codomain for the weight function. While the original lemma is often stated using integers, we selected Isabelle's natural number type (`nat`). This decision is motivated by two factors:

- **Non-negativity:** In the context of the Isolation Lemma, weight are strictly positive (from $\{1,\ldots,N\}$). Using **nat** naturally enforces this constraint without requiring additional invariants to check for negative values.
- **Inductive Reasoning:** The `nat` type in Isabelle is powerful in induction heuristics which facilitate the automation of arithmetic proofs.

HANDLING FINITENESS

The definition of `wt` utilizes Isabelle's `sum` operator(represented as $\sum$ in the math context). It is crucial to note that in Isabelle/HOL, the summation operator is only

mathematically meaningful when the set S is finite. If S were infinite, the summation would return 0 by definition in the standard library. This finiteness condition is usually implicit in informal proofs, but must be enforce explicitly in Isabelle/HOL to avoid unintended semantics.

Consequently, throughout our formalization, we do not rely on an implicit assumption that the universe in finite, which is often overlooked in pen-and-paper proofs. As it is a mandatory step in formal verification, we explicitly enforce the assumption `finite S` or `finite U` locally in all lemmas involving weight calculations. Accordingly, all lemmas involving weight computations in this development explicitly assumes `finite U` or `finite S`, making finiteness a first-class requirement rather than an implicit convention.

3.2 Defining Minimizers and Uniqueness

A key conceptual step in understanding the Isolation Lemma is indentifying what it means for a set to be a ”minimizer”. Pen-and-paper arguments often describes this idea informally – noting that some set achieves the lowest weight – but leave the structural details implicit. To move toward a formal proof, we must refine this informal notion into a logically precise predicate.

Formally, for a family of subsets $\mathscr{F}$ and a weight function w, a set A is a minimizer if:

$$A \in \mathscr{F} \quad \text{and} \quad \forall B \in \mathscr{F},\ w(A) \leq w(B) \tag{3.2}$$

To encode this concept formally, we define a predicate `minimizer F w A` that captures both membership in $\mathscr{F}$ and optimality under the weight function. This predicate holds true if and only if the set A belongs to the family F and its weight is less than or equal to the weight of any other set B in F

```
definition minimizer :: "'a set set ⇒ 'a weight ⇒'a set
   ⇒ bool" where
  "minimizer F w A =
    (A ∈ F ∧ (∀ B ∈ F. wt w A ≤ wt w B))"
```

```
  (* A is one minimizer of F under weight w *)

definition has_unique_minimizer :: "'a set set ⇒ 'a weight
    ⇒ bool" where
  "has_unique_minimizer F w =
    (∃! A ∈ F. minimizer F w A)"
  (* then A is the unique minimizer of F under weight w *)
```

Listing 3.2: Formal Definition of a Minimizer and Uniqueness

Listing 3.2 shows how the intuitive minimizer definition is directly translated into a formal predicate suitable for mechanized reasoning. It is important to note that this definition is purely predicate-based. The predicate `minimizer F w A` does not, by itself, assert the existence or uniqueness of such a set. These properties must be established separately under additional assumptions on the family F and the weight function w. In the setting of formal proof, such distinctions are neccessary to avoid implicitly assuming the existence or uniqueness of minimizers.

The Isolation Lemma goes beyond existence – its core claim concerns the **uniqueness** of the minimizer under random weights. To formalize this probabilistic uniqueness claim, we define a predicate `has_unique_minimizer`. Here, we utilize Isabelle's unique existential quantifier $\exists!$. The expression $\exists! A.P(A)$ asserts that there exists *exactly one* element A such that the predicate $P(A)$ is true. This bridges the informal idea of uniqueness with a machine-verifiable logical form.

Thus, `has_unique_minimizer F w` states that there is exactly one set A in the family F that satisfies the minimizer condition. This concise definition serves as the target statement for our main probability bound proof in the subsequent chapters.

A Useful Reformulation of Non-Uniqueness

In pen-and-paper proofs, non-uniqueness is often described informally, but in the formal setting, it is more convenient to reason with explicit witnesses. We therefore reformulate non-uniqueness using a standard logical equivalence: there is no unique

minimizer if and only if two distinct sets $A \neq B$ in $\mathscr{F}$ both satisfy the `minimizer` predicate. This reformulation enables case-based reasoning in the formal proof, where fixed witnesses simplify the structural analysis of minimizers.

3.3 The Alpha Construction

The crux of the Isolation Lemma relies on a specific value associated with each element x, commonly denoted as $\alpha(x)$. Originally introduced by Mulmuley et al. this quantity provides an intuitive threshold that governs whether x contributes to a minimum-weight set. Form an informal perspective, $\alpha(x)$ reflect how sensitive the total weight is to the choice of $w(x)$. Our goal in this section is to move from this intuition toward a formal expression that supports explicit reasoning in the proof assistant.

Mathematically, $\alpha(x)$ is defined as the difference between two minimum values:

$$\alpha(x) = \min_{S \in \mathscr{F} : x \notin S} w(S) - \min_{S \in \mathscr{F} : x \in S} w(S \setminus \{x\}) \tag{3.3}$$

Intuitively, this measures the "gap" or threshold for $w(x)$. The first term represents the minimum weight achievable without using x. The second term represents the minimum "residual" weight required to complete a set that must contain x. Taken together, these two terms describe the boundary condition under which x determines the overall minimizer.

Formal Definition

In Isabelle/HOL, we formalize this using the `LEAST` operator. The two minimum values are computed independently, and their difference is taken to obtain $\alpha(x)$.

```
definition alpha :: "('a ⇒ nat) ⇒ 'a set set ⇒ 'a ⇒ nat
   " where
  "alpha w F x =
     (LEAST n. ∃ S ∈ F. x ∉ S ∧ wt w S = n)
   - (LEAST n. ∃ S ∈ F. x ∈ S ∧ wt w (S - {x}) = n)"
```

Listing 3.3: Definition of Alpha

WELL-DEFINEDNESS AND INTENDED USE

In the transition from intuitive definitions to formal ones, a key difficulty is ensuring that expressions like $\alpha(x)$ are well-defined. The use of the `LEAST` operator in Isabelle/HOL requires explicit witnesses for the corresponding minimum conditions. This is in contrast with pen-and-paper reasoning, where such existence is often assumed without justification. In our development, such existence properties are not assumed implicitly, but are discharged later as explicit proof obligations, under suitable conditions on the family $\mathscr{F}$, in particular the existence of minimizers.

At the modeling stage, $\alpha(x)$ is treated as an abstract quantity defined in terms of minimal weights. No assumptions are made at this point about which sets realize these minimum values.

In the subsequent proof, we will show that once minimizers are known to exist, the `LEAST` expressions in the definition of $\alpha(x)$ can be simplified to concrete weight values. This separation reflects a standard discipline in formal verification, where definitions are kept independent of their required assumptions.

3.4 PROBABILITY SPACE CONSTRUCTION

We now formalize the last component required to move from intuitive probabilistic assumptions (pp) to a machine-verifiable framework (fp): the construction of the probability space. Mathematically, the Isolation Lemma assumes that the weight $w(x)$ for each element $x \in U$ is chosen independently and uniformly at random from the set $\{1,\ldots,N\}$. We assume $N \geq 1$, ensuring that the set $\{1,\ldots,N\}$ is non-empty and the uniform distribution is well-defined.

Modeling: Implicit and Explicit Probability

This stage illustrates a particularly significant shift from informal reasoning to explicit construction. In standard pen-and-paper proofs, probability is often treated implicitly. We simply state "Let w be a random weight function" and proceed to calculate probabilities of events. However, in a formal proof assistant like Isabelle/HOL, there is no ambiguity allowed. We cannot merely assume randomness; we must explicitly construct the underlying probability distribution (as a PMF object).

This introduces a gap between intuition and formalization. We must define a concrete probability mass function (PMF) over the function space $U \to \{1 \ldots N\}$. To bridge this gap, we make us of the `HOL-Probability` library, which formalizes discrete distributions using measure theory. This library makes the probability assumptions of the Isolation Lemma explicit objects of reasoning, rather than implicit conditions.

Formalizing the Product Space

To transform the intuitive statement "weights are chosen independently and uniformly" into a formal object, we construct the joint as a product of marginal distributions:

1. **Uniform Component:** A uniform distribution over the finite set $\{1, \ldots, N\}$.
2. **Product Construction:** A product PMF over the finite domain U.

In pen-and-paper reasoning, such product distributions are usually left implicit. In our formal setting, both the uniform marginals and their combination must be constructed explicitly.

The uniform marginal distribution on $\{1, \ldots, N\}$ is implemented using `pmf_of_set`, which constructs a uniform PMF over any finite non-empty set.

These marginals are then combined using the `Pi_pmf` operator, which creates a product distribution over the domain U, with independent sampling at each point.

```
definition w_pmf :: "'a set ⇒ nat ⇒ 'a weight pmf" where
  "w_pmf N U =
    Pi_pmf U (0::nat) (\<lambda>_. pmf_of_set {1..N})"
```

```
(* The probability mass function for random weights *)
```

Listing 3.4: Construction of the Weight Probability Space

As shown in Listing 3.4, `w_pmf N U` defines the probability measure over total functions of type `'a => nat`, where the random choices are made independently on U. The construction uses the `Pi_pmf` operator (mathematically Π) to combine individual uniform distributions into a single product measure. This explicitly captures the **independence** assumption, ensuring that the choice of weight for any $x \in U$ does not influence others.

A technical nuance in this definition is the argument `(0::nat)` passed to `Pi_pmf`. Since functions in Isabelle/HOL are total, they must return a value even for inputs outside their logical domain ($x \notin U$). Here, `0` serves as the default value for such out-of-domain queries. While this implementation detail is necessary to satisfy Isabelle's type system, it does not affect the mathematical validity of the lemma, as our reasoning is strictly confined to the elements within U.

By modeling the randomness as a PMF object, probabilistic events become first-class objects in the logic, allowing standard measure theoretic reasnong, such as union bounds, to be carried out within Isabelle/HOL.

From Modeling to Proof. With the modeling phase complete, we now have a precise formal infrastructure: weight assignments over finite universes, a definition of minimizers and uniqueness, and an explicitly constructed product probability space.

This framework resolves the ambiguities present in informal (pen-and-paper) reasoning, where independence, randomness, and uniqueness are often stated but not justified.

In the next chapter, we transit from formal modeling to fully formal verification. Each proof step will rely on the definitions developed here and will make all logical dependencies explicit, thus completing the transition from intuitive understanding to machine-checked formal proof.

4 PROOF CONSTRUCTION

In this chapter, we present the formal proof of the Isolation Lemma. Unlike pen-and-paper proofs, formal verification requires every mathematical step to be made fully explicit. This includes not only the main reasoning steps, but also background obligations such as existential witnesses, measurability, independence, type correctness, and totality.

Our goal is to construct a fully formal proof (FP) of the Isolation Lemma in Isabelle/HOL. However, it is neither practical nor desirable to translate a pen-and-paper (PP) proof directly into formal code. Instead, we introduce an intermediate stage, which we refer to as the *precise pen-and-paper* (PPP) level. This stage consists of structurally explicit arguments and decompositions that closely resemble formal code, but are still described in human-readable mathematical language.

Thus, each proof component in this chapter is presented in three layers:

- **PP:** The intuitive pen-and-paper ideas from the literatures.
- **PPP:** A human-readable structured and detailed decomposition of the argument, refined to match the constraints of the formal system.
- **FP:** The final formalization codes in Isabelle/HOL.

We organize the whole proof as a sequence of structural reductions: We begin by establishing the existence of minimizers and by reformulating its non-uniqueness in a constructive form. Then we analyze the auxiliary quantity $\alpha(x)$, which plays a central role in isolating the dependence on individual weights. This leads to a reduction of the uniqueness failure to an explicit collision event of the form $w(x) = \alpha(x)$. Finally, we bound the probability of such events using the product probability space constructed in Chapter 3.

4.1 Existence of Minimizers

Before analyzing the uniqueness of minimizers, we must first ensure that the notion of a minimizer is well-defined. In particular, for a given family of sets $\mathscr{F}$ and a weight function w, it is necessary to establish that at least one minimizer exists whenever $\mathscr{F}$ is non-empty.

PP. In informal proofs, this step is usually omitted. Since weights are natural number and the family $\mathscr{F}$ is assumed to be non-empty, the existence of a minimizer is considered self-evident.

PPP. In preparation for formalization, however, we must isolate this assumption and state it precisely, since later constructs (like $\alpha(x)$ or uniqueness predicates) depend on the existence of at least one such set.

Lemma (Existence of a Minimizer). *If $\mathscr{F} \neq \emptyset$, then there exists a set $A \in \mathscr{F}$ such that A is a minimizer of $\mathscr{F}$ under the weight function w.*

FP. This lemma is formalized as:

```
lemma exist_minimizer:
  assumes "F ≠ {}"
  obtains A where "A ∈ F" "minimizer F w A"
```

Listing 4.1: Existence of Minimizer

The proof proceeds by constructing the minimal weight value using Isabelle's `LEAST` operator. First, we extract a witness $S_0 \in \mathscr{F}$ to ensure the minimum is well-defined. We then define the minimal weight m as the least n such that some $S \in \mathscr{F}$ satisfies $wt\ w\ S = n$. A corresponding set A is then chosen to realize this minimum. Finally, we show that this A has weight at most that any other set in $\mathscr{F}$, thus satisfying the `minimizer` predicate introduced in Chapter 3.

This lemma plays a foundational role: it ensures that future uses of the `LEAST` operator or comparisons among minimizers are grounded in an actual, verifiable witness. By

discharging this assumption with a lemma, subsequent lemmas can avoid repeatedly re-establishing this basic existence condition.

4.2 REFORMULATING THE NON-UNIQUENESS

The Isolation Lemma asserts that with high probability, the weight function w yields a unique minimizer in $\mathscr{F}$. To prove this in a formal setting, we must first address the failure of uniqueness – i.e., the possibility that multiple sets in $\mathscr{F}$ achieve the minimum weight.

PP. In pen-and-paper proofs, non-uniqueness is usually described in a purely intuitive way: saying that a minimizer is ”not unique” simply means that there exist at least two different sets in $\mathscr{F}$ that both attain the minimum weight. This idea is often expressed informally by statements such as ”there are two distinct minimizers”, without any further logical unpacking.

PPP. To make this intuition compatible with formal reasoning,we refine it into a positive existential formulation. Instead of negating a uniqueness statement, we characterize failure of uniqueness by the explicit existence of two distinct witnesses that both satisfy the minimizer predicate.

Concretely, the minimizer of $\mathscr{F}$ under w is not unique if and only if there exist two distinct sets $A \neq B$ in $\mathscr{F}$ such that both are minimizers. In logical terms, this corresponds to replacing a negated unique-existence statement by an equivalent existential condition of the form

$$\exists A\, B.\, A \in \mathscr{F} \;\wedge\; B \in \mathscr{F} \;\wedge\; \text{minimizer}(\mathscr{F}, w, A) \;\wedge\; \text{minimizer}(\mathscr{F}, w, B) \;\wedge\; A \neq B.$$

This reformulation eliminates the outer negation and transformes the non-uniqueness into concrete objects that can be fixed in subsequent arguments.

Moreover, if A and B are both minimizers, then they must attain the same total weight. This follows directly from the definition of the minimizer predicate: each has weight at

most that of any other set in $\mathscr{F}$, and hence in particular at most that of the other. This elementary but explicit observation will be used later when comparing the contribution of individual elements in two different minimizers.

Lemma (Non-Uniqueness via Two Minimizers). *Assume $\mathscr{F} \neq \emptyset$. Then the minimizer of $\mathscr{F}$ under w is not unique if and only if there exist two distinct sets $A, B \in \mathscr{F}$ such that both A and B are minimizers; equivalently, there exist $A \neq B$ in $\mathscr{F}$ with*

$$\forall S \in \mathscr{F},\ w(A) \leq w(S) \ \wedge\ w(B) \leq w(S),$$

and in particular $w(A) = w(B)$.

FP. These facts are formalized as:

```
lemma not_unique_iff_exist_two:
  assumes "F ≠ {}"
  shows "¬ has_unique_minimizer F w ⟷
        (∃ A B. A ∈ F ∧ B ∈ F ∧
                minimizer F w A ∧
                minimizer F w B ∧
                A ≠ B)"
```

Listing 4.2: Non-Uniqueness Characterization

```
lemma minimizers_same_weight:
  assumes "minimizer F w A" "minimizer F w B"
  shows "wt w A = wt w B"
```

Listing 4.3: Minimizers Have Equal Weight

This reformulation replaces a negated uniqueness predicate by a positive existential statement that provides concrete witnesses. Once such witnesses A and B are fixed, later arguments can reason about their structural relationship, for example by analyzing elements in the symmetric difference $A \triangle B$. This will be a key step in the collision characterization developed in subsequent sections.

As an immediate consequence of this characterization, if A and B are both minimizers, they must attain the same total weight. This follows directly from the definition of the minimizer predicate and will be used implicitly in later comparisons between different minimizers.

4.3 THE ALPHA CONSTRUCTION

A central ingredient in the proof of the Isolation Lemma is the auxiliary quantity $\alpha(x)$ associated with each element $x \in U$. Intuitively, $\alpha(x)$ isolates the dependence of the total weight of a set on the individual weight $w(x)$, allowing us to reason about the effect of modifying a single coordinate.

Formally, $\alpha(x)$ is defined as the difference between two minimal weight values determined by the family $\mathscr{F}$. The first term represents the minimum total weight among all sets in $\mathscr{F}$ that do not contain x. The second term represents the minimum residual weight required to form a set in $\mathscr{F}$ that contains x, excluding the contribution of x itself. This definition was introduced by Mulmuley et al. and plays a crucial role in isolating single-coorrdinate events.

In our formalization, $\alpha(x)$ is defined using Isabelle's `LEAST` operator, which selects the least natural number satisfying a given predicate. At this stage, $\alpha(x)$ is treated as an abstract quantity defined in terms of minimal weights, without assuming that the minima are realized by any particular sets.

In the present chapter, we analyze its properties and show how the abstract `LEAST`-based definition can be simplified in subsequent arguments, once the existence of minimizers has been derived.

4.3.1 SIMPLIFYING THE DEFINITION OF $\alpha(x)$

PP. In pen-and-paper proofs, the two minimization terms in the definition of $\alpha(x)$ are usually treated as concrete minimum values realized by suitable sets in $\mathscr{F}$. Once a minimizer exists, the least total weight among sets not containing x is simply the weight of a minimizer that avoids x. Likewise, the least residual weight among sets

containing x is obtained by removing x from a minimizer that contains it. This allows the abstract LEAST-based definition of $\alpha(x)$ to be replaced by an ordinary arithmetic expression involving concrete minimizers.

PPP. To make this intuition precise, we analyze the two LEAST expressions in the definition of $\alpha(x)$ separately.

First, suppose that B is a minimizer of $\mathscr{F}$ under the weight function w, and that $x \notin B$. Since $B \in \mathscr{F}$ and attains the minimum total weight among all sets in $\mathscr{F}$, it satisfies the defining predicate of the first LEAST expression. Moreover, any other set $S \in \mathscr{F}$ with $x \notin S$ must have weight at least $w(B)$. Hence, the minimum total weight among all sets in $\mathscr{F}$ that do not contain x is attained by B, and the first LEAST expression can be shown equal to the concrete value $W(B)$.

Second, suppose that A is a minimizer of $\mathscr{F}$ and that $x \in A$. Removing x from A yields a residual set whose total weight equals $w(A) - w(x)$. Since $A \in \mathscr{F}$ and attains the minimum total weight among all sets in $\mathscr{F}$, any other set $S \in \mathscr{F}$ containing x must have residual weight at least $w(A \setminus \{x\})$. Therefore, the least value selected by the second LEAST operator coincides with the concrete value $w(A \setminus \{x\})$.

Together, these two observations eliminate both instances of the LEAST operator from the definition of $\alpha(x)$ whenever suitable minimizers are available. They reduce the abstract definition to a concrete arithmetic form that can be used directly in subsequent arguments.

FP. These simplifications are formalized by the following auxiliary lemmas in Isabelle/HOL:

```
lemma minimizer_B_eq_least_not_in:
  assumes "minimizer F w B" "x ∉ B"
  shows "(LEAST n. ∃ S. S ∈ F ∧ x ∉ S ∧ wt w S = n)
         = wt w B"
```

Listing 4.4: First LEAST Eliminated by a Minimizer

```
lemma minimizer_A_eq_least_in_residual:
```

```
  assumes "finite U" "F ⊆ Pow U" "minimizer F w A" "x ∈ A
    "
  shows "(LEAST n. ∃ S. S ∈ F ∧ x ∈ S ∧ wt w (S - {x}) = n)
        = wt w (A - {x})"
```

Listing 4.5: Second LEAST Eliminated by a Minimizer

These lemmas connect the abstract `LEAST` expressions in the definition of $\alpha(x)$ with concrete weights of minimizers. In particular, they justify replacing both minimization terms by explicit arithmetic values whenever a minimizer avoiding x or containing x is available.

As a result, the auxiliary quantity $\alpha(x)$ can be rewritten in terms of ordinary arithmetic expressions, which will be essential for proving its structural properties and for carrying out the probabilistic analysis in later sections.

4.3.2 Independence of $\alpha(x)$ from Individual Weights

PP. Intuitively, $\alpha(x)$ should not depend on the particular value of the single coordinate $w(x)$. Changing the value of $w(x)$ shifts the total weight of every set containing x by the same offset, while leaving the relative ordering of weights determined by the remaining elements unchanged. Hence the two minima defining $\alpha(x)$ – one over sets excluding x, and one over residual weights of sets including x – remain invariant under modifications of $w(x)$.

PPP. We now formulate this independence property as a formal lemma, which will serve as a key technical interface for the probabilistic argument in the next section.

Lemma (Independence of $\alpha(x)$). *Let w and w' be two weight functions such that $w(y) = w'(y)$ for all $y \in U \setminus \{x\}$. Then $\alpha(x)$ computed with respect to w is equal to $\alpha(x)$ computed with respect to w'.*

To prove this lemma, we compare the two definitions of $\alpha(x)$ under w and w'. The core observation is that for any set $S \subseteq U$ with $x \notin S$, the total weight $wt\ w\ S$ depends only on values of w on $S \subseteq U \setminus \{x\}$, and therefore equals $wt\ w'$§. The same reasoning

applies to residual sets of the form $S - \{x\}$, since $S - \{x\} \subseteq U \setminus \{x\}$.

Using this congruence of weights, we show that the predicates appearing in the two LEAST expressions defining $\alpha(x)$ are equivalent under w and w'.

$$\exists S \in F.\, x \notin S \wedge wt\ w\ S = n \quad \Longleftrightarrow \quad \exists S \in F.\, x \notin S \wedge wt\ w'\ S = n,$$

and similarly for the residual term

$$\exists S \in F.\, x \in S \wedge wt\ w\ (S - \{x\}) = n \quad \Longleftrightarrow \quad \exists S \in F.\, x \in S \wedge wt\ w'\ (S - \{x\}) = n.$$

Therefore, the two LEAST values coincide under w and w'. Unfolding the definition of $\alpha(x)$ then yields the desired conclusion of the lemma, namely $\alpha_w(x) = \alpha_{w'}(x)$.

FP. This lemma is formalized as `alpha_independent_of_coord` in Isabelle/HOL:

```
lemma alpha_independent_of_coord:
  assumes "finite U" "F ⊆ Pow U" "x ∈ U"
  shows "∀ w w'. (∀ y. y ∈ U - {x} ⟶ w y = w' y)
        ⟶ alpha (w :: 'a ⇒ nat) F x = alpha w' F x"
```

Listing 4.6: Independence of alpha(x) from the x-coordinate

This lemma is proved by establishing weight congruence on sets contained in $U - \{x\}$, transferring this congruence to the defining predicates of the two LEAST expressions in $\alpha(x)$, and finally unfolding `alpha_def`.

4.3.3 Collision characterization via $\alpha(x)$

PP. We now establish the key structural link between the failure of uniqueness and the auxiliary quantity $\alpha(x)$.

Assume that the minimizer of $\mathscr{F}$ under w is not unique. Then there exist two distance minimizers $A, B \in \mathscr{F}$. Choose an element $x \in A \setminus B$. Since A and B are minimizers, their total weights coincide:

$$w(A) = w(B).$$

Moreover, in the definition of $\alpha(x)$, the minimum over sets not containing x is realized by B, while the minimum over residual weights of sets containing x is realized by $A \setminus \{x\}$. Hence,

$$\alpha(x) = w(B) - w(A \setminus \{x\}) = w(A) - \big(w(A) - w(x)\big) = w(x).$$

Therefore, non-uniqueness implies the existence of an element $x \in U$ such that $w(x) = \alpha(x)$.

PPP. The pen-and-paper argument above already matches closely the structure of the formal proof. The only additional ingredients required for formalization are the explicit side conditions ensuring that the two `LEAST` expression in the definitions of $\alpha(x)$ are well-defined and can be eliminated.

Concretely, we assume that the universe U is finite and that the family $\mathscr{F}$ is a collection of subsets of U. Under these assumptions, the two auxiliary lemmas established in Section 4.3 apply: the first shows that the `LEAST` term over sets not containing x is realized by B whenever $x \notin B$, and the second shows that the residual `LEAST` term over sets containing x is realized by $A \setminus \{x\}$ whenever $x \in A$.

Together with the lemma stating that any two minimizers have equal total weight, these results allow the formal proof to mirror the pen-and-paper derivation step by step. No additional combinatorial insight is required at this stage; the formalization consists entirely in assembling previously established lemmas and discharging the necessary finiteness and membership conditions.

FP. This reasoning is captured in Isabelle/HOL by the following lemma:

```
lemma two_minimizers_alpha_eq:
  assumes "finite U" "F ⊆ Pow U"
          "A ∈ F" "B ∈ F"
          "minimizer F w A" "minimizer F w B"
          "x ∈ A - B"
  shows   "alpha w F x = w x"
```

Listing 4.7: Collision Witnessed by a Single Coordinate

The proof unfolds the definition of $\alpha(x)$ and eliminates both `LEAST` expressions using the auxiliary lemmas from Section 4.3. The equality of weights $w(A) = w(B)$ follows from the fact that A and B are both minimizers. Finally, the standard splitting identity $w(A) = w(A \setminus \{x\}) + w(x)$ is used to derive the desired conclusion $\alpha(x) = w(x)$.

4.4 Probabilistic Analysis and Final Bound

PP. We now complete the proof of the Isolation Lemma by bounding the probability that the minimizer of $\mathscr{F}$ is not unique under a random weight assignment. Throughout this section, weights are sampled according to the product probability space constructed in Chapter 3.

By the collision characterization established in Section 4.3.3, the event that the minimizer is not unique can be reduced to a union of single-coordinate events:

$$\{\,\text{the minimizer is not unique}\,\} \subseteq \bigcup_{x \in U} \{\, w(x) = \alpha(x) \,\}.$$

Thus, it suffices to bound the probability of each event $\{\, w(x) = \alpha(x) \,\}$.

Fix an arbitrary element $x \in U$. By the independence property proved in Section 4.3.2, the auxiliary quantity $\alpha(x)$ depends only on the weights assigned to elements in $U \setminus \{x\}$. Consequently, once all other weights have been fixed, $\alpha(x)$ becomes a constant with respect to the remaining random choice of $w(x)$.

Under the product probability space, the random variable $w(x)$ is distributed uniformly over the finite set $\{1, \ldots, N\}$ and is independent of all other coordinates. Therefore, the probability that $w(x)$ attains any particular value is exactly $1/N$, and in particular

$$\Pr\big[\, w(x) = \alpha(x) \,\big] \leq \frac{1}{N}.$$

Since the universe U is finite, we may apply the union bound:

$$\Pr\big[\text{the minimizer is not unique}\big] \leq \sum_{x \in U} \Pr\big[\, w(x) = \alpha(x) \,\big] \leq \frac{|U|}{N}.$$

Equivalently, the probability that the minimizer is unique satisfies

$$\Pr\big[\text{the minimizer is unique}\big] \geq 1 - \frac{|U|}{N}.$$

This inequality constitutes the statement of the Isolation Lemma.

PPP. The formal proof follows the same structure as the pen-and-paper argument, but it requires an explicit decomposition of the probabilistic reasoning into measurable events and independent random variables.

First, the collision event $\{\, w(x) = \alpha(x) \,\}$ is rewritten as a finite disjunction:

$$\{\, w(x) = \alpha(x) \,\} \;=\; \bigcup_{k=1}^{N} \{\, \alpha(x) = k \;\wedge\; w(x) = k \,\}.$$

This step relies on the fact that under the product probability space, the coordinate random variable $w(x)$ almost surely takes values in $\{1, \ldots, N\}$, while all other values have probability zero.

Second, to obtain a multiplicative bound on each summand, the independence between $w(x)$ and $\alpha(x)$ must be established at the level of random variables. This is achieved by exploiting the product structure of the underlying PMF. The restriction of a sample function to $\{x\}$ and to $U \setminus \{x\}$ yields two independent components. Since $w(x)$ depends only on the first component, while $\alpha(x)$ depends only on the second (by the independence-of-α lemma), their compositions are independent random variables. This yields

$$\Pr[\, \alpha(x) = k \wedge w(x) = k \,] = \Pr[\, \alpha(x) = k \,] \cdot \Pr[\, w(x) = k \,].$$

Finally, the construction of $\alpha(x)$ via minimum operators over finite families requires

auxiliary measurability arguments. Finite sums of coordinates are shown to be measurable, predicates of the form $\exists S \in F. \cdots$ are measurable over the product space, and the resulting LEAST expressions preserve measurability. These facts ensure that all events appearing in the above decomposition are measurable and that the standard independence and union-bound lemmas can be applied without further side conditions.

FP. In Isabelle/HOL, the above structural steps are realized by the following core lemmas and theorems:

(a) Coordinate support and point probabilities. We first show that for any $x \in U$, the random variable $w(x)$ takes values almost surely in $\{1,\ldots,N\}$. This yields the following support lemma:

```
lemma w_pmf_component_in_range:
  assumes "x ∈ U" "N ≥ 1" "finite U"
  shows "set_pmf (map_pmf (λ w. w x) (w_pmf N U)) ⊆ {1..N
    }"
```

Listing 4.8: Coordinate Support

Based on this, we derive the uniform point probability formula for each $k \in \{1,\ldots,N\}$ (appearing as a subgoal in the main proof):

```
measure_pmf.prob (w_pmf N U) {w. w x = k} = 1 / real N
```

Listing 4.9: Point Probability of One Coordinate

(b) Measurability lemmas (measurability of α). To legitimately apply probabilistic reasoning, we must show that the auxiliary quantity $\alpha(x)$ is a measurable random variable. The formalization culminates in:

```
lemma alpha_measurable:
  assumes "finite U" "F ⊆ Pow U" "x ∈ U"
  shows "(λ f. alpha (λ y. if y ∈ U-{x} then f y else 0) F
    x)
```

```
        ∈ (Π_M (U-{x}) (λ _. count_space UNIV)) →_M (
          count_space UNIV)"
```

Listing 4.10: Measurability of Alpha

This proof relies on measurability of finite sums, measurability of finite existential predicates, and the measurability theorem for the `LEAST` operator.

(c) Independence lemma. Next, we lift coordinate-wise independence to random-variable independence by showing that $w(x)$ is independent of $\alpha(x)$, which depends only on the remaining coordinates:

```
prob_space.indep_var (measure_pmf (w_pmf N U))
  (count_space UNIV) (λ w. w x)
  (count_space UNIV) (λ w. alpha w F x)
```

Listing 4.11: Independence of Coordinate and Alpha

This step combines restriction lemmas for product measures with the previously established lemma `alpha_independent_of_coord`, ensuring that $\alpha(x)$ depends solely on the weights assigned to $U - \{x\}$.

(d) Main theorem: Isolation Lemma. Finally, combining the single-coordinate probability bound with the union bound yields the formalized version of the Isolation Lemma:

```
theorem isolation_lemma_main:
  fixes U :: "'a set" and F :: "'a set set" and N :: nat
  assumes "finite U" "F ⊆ Pow U" "F ¬eq {}" "N ≥ 1"
  shows "measure_pmf.prob (w_pmf N U){w.
     has_unique_minimizer F w}
        ≥ 1 - real(card U) / real N"
```

Listing 4.12: Main Theorem: Isolation Lemma

This theorem completes the formal derivation from the structural collision event to the global lower bound on the probability of uniqueness.

Taken together, these lemmas bridge the gap between the informal probabilistic argument and its fully formal counterpart. They make explicit the two technical ingredients that are implicit in pen-and-paper proofs: the measurability of $alpha(x)$ and its independence from the coordinate $w(x)$. With these in place, the Isolation Lemma follows directly by applying the union bound inside Isabelle/HOL's probability framework.

CHAPTER SUMMARY

In this chapter, we presented a fully formal proof of the Isolation Lemma in Isabelle/HOL, transforming Pen-and-Paper Proof into computer-verifiable Formal Proof. We reformulated non-uniqueness as the existence of two distinct minimizers, introduced the auxiliary quantity $\alpha(x)$, and showed how it captures the precise collision event responsible for ties. A key technical contribution was the proof that $\alpha(x)$ is independent of the individual weight $w(x)$, enabling a reduction to single-coordinate probabilistic events. Finally, by establishing measurability, independence, and a uniform point probability bound within a product probability space, we derived the standard $1 - |U|/N$ uniqueness bound in a fully mechanized manner.

5 Perfect Matching Problem

TODO: Formal proof of Parallel Algorithm for Perfect Matching Problem.

Note: Formal proof of this algorithm has been fully implemented and pushed to github repository; however, code cleaning is still needed. I will work on cleaning the code in the following weeks and document the finalized implementation after that.

6 CONCLUSION AND FUTURE WORK

REFERENCES

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